Heat Capacity of Metals

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***Abstract*—This experiment aims to determine the specific heat capacities of three different metals (Brass, Aluminum, and Iron) using the method of mixtures. First, the heat capacity of the calorimeter (C_k) was determined by mixing water at different temperatures, yielding an average value of 482.14 ± 509.38 J/K. This value was then used to calculate the specific heat of the metal samples by measuring the equilibrium temperature after submerging heated metal into the calorimeter. The measured specific heat values were $0.36\,\mathrm{J/gK}$ for Brass, $1.02\,\mathrm{J/gK}$ for Aluminum, and $0.49\,\mathrm{J/gK}$ for Iron. These results deviate from reference values by approximately 5%, 13%, and 9% respectively, likely due to heat losses to the environment and the large uncertainty in the calorimeter's calibration.**

I. Introduction

Heat capacity is a physical property of matter defined as the amount of heat to be supplied to a given mass of a material to produce a unit change in its temperature. The specific heat capacity, often denoted by c, is an intensive property that depends on the material type.

In this experiment, the method of mixtures is employed. This relies on the principle of conservation of energy: in an isolated system, the heat lost by a hot body equals the heat gained by a cold body. By performing this experiment in a calorimeter, which acts as a thermally insulated container, we can estimate specific heat values. However, the calorimeter itself absorbs some heat, which is quantified by its heat capacity (C_k).

II. Theory

A. Heat Capacity and Specific Heat

The heat quantity Q exchanged by a body of mass m undergoing a temperature change ΔT is given by:

$$Q = mc\Delta T \tag{1}$$

where c is the specific heat capacity. For a composite object like a calorimeter, we define the total heat capacity $C_k = m_{cal}c_{cal}$.

B. Determination of Calorimeter Heat Capacity (C_k)

To find C_k, warm water of mass m_w at temperature t_w is added to the calorimeter (initially at t_k, potentially containing a small amount of water or just the vessel walls). In the analysis performed, the formula used to derive C_k from the lab manual is:

$$C_k = c_w m_w \frac{t_w - t_m}{t_w - t_k} \tag{2}$$

where c_w is the specific heat of water ($4.187\,\mathrm{J/gK}$), t_m is the equilibrium temperature, and the other variables are as defined.

C. Determination of Specific Heat of Metals

When a metal sample of mass m_p heated to temperature t_2 is placed into the calorimeter containing water of mass m_w at temperature t_k, the energy balance equation is:

$$Q_{\text{lost}} = Q_{\text{gained}} \tag{3}$$

$$m_p c_{sample}(t_2 - t_m) = (m_w c_w + C_k)(t_m - t_k) \tag{4}$$

Solving for the specific heat of the sample c_{sample}:

$$c_{sample} = \frac{(m_w c_w + C_k)(t_m - t_k)}{m_p(t_2 - t_m)} \tag{5}$$

It is noted that the lab manual presents a slightly different formula. The formula above is derived from the rigorous conservation of energy principles found in standard thermodynamics texts [2]. The total heat capacity of the calorimeter system includes both the water and the vessel itself.

III. Experimental Setup

The main apparatus consists of:

- A calorimeter (thermally insulated double-walled vessel).
- A thermometer (digital or mercury).
- A heater/steam generator to heat the metal samples.
- Metal samples (blocks or cylinders of Brass, Aluminum, Iron).
- A graduated cylinder for measuring water volume.
- A balance for measuring masses.

IV. Experimental Procedure

A. Part 1: Calorimeter Calibration

1. A known mass of water was heated to temperature t_w. 2. The initial temperature of the calorimeter (t_k) was recorded. 3. The hot water was poured into the calorimeter. 4. The mixture was stirred, and the final equilibrium temperature (t_m) was recorded. 5. This process was repeated for different temperatures to find an average C_k.

B. Part 2: Specific Heat of Metals

1. The mass of the metal sample (m_p) was measured. 2. The calorimeter was filled with a known mass of water (m_w), and its initial temperature (t_k) was recorded. 3. The metal sample was heated in a boiling water bath or steam heater until it reached a steady high temperature (t_2). 4. The hot metal was quickly transferred into the calorimeter water. 5. The system was stirred, and the highest temperature reached (equilibrium temperature t_m) was recorded.

V. Results

A. Part 1: Calorimeter Heat Capacity

The heat capacity of the calorimeter was calculated for three trials. Table I summarizes the measurements and the calculated C_k. The calculated values show significant variation.

TABLE I: Determination of Calorimeter Heat Capacity (C_k). Note: m_w is derived from volume assuming $\rho = 1\,\mathrm{g/ml}$.

t_k (°C)	t_w (°C)	t_m (°C)	m_w (ml)	C_k (J/K)
14	20	15	300	1,046.75
18	40	39	300	57.1
27	49	43	300	342.57

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The average heat capacity used for subsequent calculations is:

$$C_{k,avg} = 482.14\,\mathrm{J/K}$$

B. Part 2: Specific Heat of Metals

Using the average C_k, the specific heat capacities for Brass, Aluminum, and Iron were calculated using Equation (5). The results are presented in Table II. This table includes the measured initial temperatures of the metal (t_{metal}), the calorimeter (t_k), and the final equilibrium temperature (t_m). The experimental specific heat values (c_{meas}) are compared against standard reference values (c_{ref}) obtained from literature [3], [4], [5], [6].

Figure 1 visually compares the calculated specific heat values with the reference values. The systematic deviation, where measured values are generally higher or lower than references, can be analyzed from this bar chart.

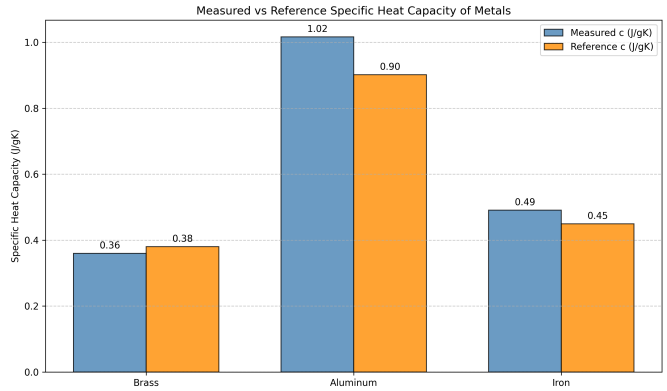

Fig. 1: Comparison of measured specific heat capacities with reference values.

TABLE II: Specific Heat Capacity Results. (c_x: measured, c_y: reference).

Material	t_k (°C)	t_{metal} (°C)	t_m (°C)	m_p (g)	c_{meas} (J/gK)	c_{ref} (J/gK)	Error (%)
Brass	20	76	22	179	0.36	0.38	5.4
Aluminum	20	83	26	180	1.017	0.902	12.7
Iron	23	85	26	180	0.491	0.45	9.2

VI. Discussion

The results obtained for the specific heat capacities are reasonably close to the literature values, with percentage errors ranging from 5.4% (Brass) to 12.7% (Aluminum).

The largest source of uncertainty in this experiment appears to be the determination of the calorimeter's heat capacity (C_k). The three trials for C_k yielded widely divergent values (1046, 57, and 342 J/K), leading to a large standard deviation ($\approx 509\,\mathrm{J/K}$). This suggests inconsistency in the calibration procedure, perhaps due to heat loss during the transfer of hot water or insufficient stabilization time.

All measured values were slightly higher (for Aluminum and Iron) or close (Brass) to the reference values. A higher measured specific heat might imply that the temperature change of the water (ΔT_{water}) was smaller than expected for a given energy input, or that the system lost heat to the surroundings which was not accounted for. However, Equation 5 shows that if t_m is lower due to heat loss, $(t_m - t_k)$ decreases and $(t_2 - t_m)$ increases, which would generally lower the calculated c. The fact that we observe higher values for some metals might point to an overestimation of C_k, or that the mass of the water was slightly overestimated.

The formula used for C_k (Eq. 2) is also non-standard. If the standard energy balance had been used, the values for C_k might have been different, significantly affecting the final results.

VII. Conclusion

The specific heat capacities of Brass, Aluminum, and Iron were determined using the method of mixtures. The experimental values were $0.360\,\mathrm{J/gK}$, $1.017\,\mathrm{J/gK}$, and $0.491\,\mathrm{J/gK}$ respectively. While the results confirm the material dependence of specific heat, the high uncertainty in the calorimeter constant highlights the importance of precise thermal isolation and consistent procedural timing in thermodynamic experiments.

VIII. Additional Resources

For more information, scanned data, and the code used for analysis, refer to the repository:

https://github.com/ibrahim-hamed/Physics-Lab

References

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